```
>> dynare mcmc.mod
Starting Dynare (version 6.4).
Calling Dynare with arguments: none
Starting preprocessing of the model file ...
Found 25 equation(s).
Evaluating expressions...
Computing static model derivatives (order 2).
Computing static model derivatives w.r.t. parameters (order 2).
Normalizing the static model...
Finding the optimal block decomposition of the static model...
7 block(s) found:
  5 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                               and 16 feedback variable(s).
Computing dynamic model derivatives (order 2).
Computing dynamic model derivatives w.r.t. parameters (order 2).
Normalizing the dynamic model...
Finding the optimal block decomposition of the dynamic model...
5 block(s) found:
  3 recursive block(s) and 2 simultaneous block(s).
  the largest simultaneous block has 16 equation(s)
                               and 16 feedback variable(s).
Preprocessing completed.
Preprocessing time: 0h00m00s.

STEADY-STATE RESULTS:

q                29.8013
k                1.46248
R                1.01516
Rp               1.0121
b                42.9331
x                0.0137035
varphi           0.0317528
mu               0.00522869
kp               2.67752
chi              1.02332
nu               0.00416305
eta              1.38229
zeta             1.02332
N                11.7043
phi              0.272618
Ne               11.6419
Nn               0.0624575
xp               2.27908
bp               31.2288
Y                2.29278
eb               0
eN               0
eY               0
N_obs            0
Y_obs            0

EIGENVALUES:
         Modulus             Real        Imaginary
               0                0                0
       5.563e-18        5.563e-18                0
       2.813e-17        2.813e-17                0
```

```
        3.86e-15       -3.86e-15               0
       1.442e-14      -1.442e-14               0
       1.564e-14       1.564e-14               0
       1.442e-12      -1.442e-12               0
       1.444e-12       1.444e-12               0
         0.01476         0.01476               0
             0.8             0.8               0
             0.8             0.8               0
          0.9871          0.9871               0
           1.016           1.016               0
            1.02            1.02               0
           1.021           1.021               0
           1.022           1.022               0
       2.131e+14      -2.131e+14               0
       8.043e+17       8.043e+17               0
             Inf             Inf               0

There are 7 eigenvalue(s) larger than 1 in modulus for 7 forward-looking variable(s)
The rank condition is verified.


======== Identification Analysis ========

Testing prior mean

Note that differences in the criteria could be due to numerical settings,
numerical errors or the method used to find problematic parameter sets.
Settings:
    Derivation mode for Jacobians:                         Analytic using sylvester equations
    Method to find problematic parameters:                 Nullspace and multicorrelation
coefficients
    Normalize Jacobians:                                   Yes
    Tolerance level for rank computations:                 robust
    Tolerance level for selecting nonzero columns:         1e-08
    Tolerance level for selecting nonzero singular values: 1e-03

REDUCED-FORM:
    All parameters are identified in the Jacobian of steady state and reduced-form solution
matrices (rank(Tau) is full with tol = robust).

MINIMAL SYSTEM (KOMUNJER AND NG, 2011):
    All parameters are identified in the Jacobian of steady state and minimal system
(rank(Deltabar) is full with tol = robust).

SPECTRUM (QU AND TKACHENKO, 2012):
    All parameters are identified in the Jacobian of mean and spectrum (rank(Gbar) is full with
tol = robust).

MOMENTS (ISKREV, 2010):
    All parameters are identified in the Jacobian of first two moments (rank(J) is full with tol
= robust).

==== Identification analysis completed ====


You did not declare endogenous variables after the estimation/calib_smoother command.
Initial value of the log posterior (or likelihood): -161450.1458
-----------------
```

```
f at the beginning of new iteration,    161450.1457763030
Predicted improvement: 48966083544.640724182
lambda =          1; f =      9925878.7504030
lambda =    0.33333; f =      1240729.7125148
lambda =    0.11111; f =       280038.0094950
lambda =   0.037037; f =       174254.1371084
lambda =   0.012346; f =       162754.5256493
lambda =  0.0041152; f =       161559.1793328
lambda =  0.0013717; f =       161453.6945468
lambda = 0.00045725; f =           61.3879795
lambda = 0.00015242; f =          447.1928296
lambda = 5.0805e-05; f =         2268.0892086
lambda = 1.6935e-05; f =         7351.6034955
lambda =  5.645e-06; f =        24834.2178761
lambda = 1.8817e-06; f =        66777.5563330
Norm of dx     3129.4
----
Improvement on iteration 1 =   161388.757796821
-----------------
f at the beginning of new iteration,        61.3879794820
Predicted improvement:       47.786639694
lambda =          1; f =           61.3723869
lambda =    0.33333; f =           50.3747815
Norm of dx   0.097762
----
Improvement on iteration 2 =       11.013197985
-----------------
f at the beginning of new iteration,        50.3747814972
Predicted improvement:        1.699440898
lambda =          1; f =           47.1339566
lambda =     1.9332; f =           44.3656315
lambda =     3.7372; f =           39.6402473
lambda =     7.2247; f =           33.3352234
Norm of dx   0.018668
----
Improvement on iteration 3 =       17.039558130
-----------------
f at the beginning of new iteration,        33.3352233676
Predicted improvement:        3.181756793
lambda =          1; f =           29.2294820
Norm of dx     0.0787
----
Improvement on iteration 4 =        4.105741336
-----------------
f at the beginning of new iteration,        29.2294820312
Predicted improvement:        6.004759490
lambda =          1; f =           22.3975701
Norm of dx    0.12013
----
Improvement on iteration 5 =        6.831911915
-----------------
f at the beginning of new iteration,        22.3975701162
Predicted improvement:        1.146015275
lambda =          1; f =           20.7752767
lambda =     1.9332; f =           20.4044957
Norm of dx   0.085967
----
Improvement on iteration 6 =        1.993074412
```

```
-----------------
f at the beginning of new iteration,        20.4044957038
Predicted improvement:        0.237353134
lambda =          1; f =           20.1267924
Norm of dx  0.0043401
----
Improvement on iteration 7 =        0.277703285
-----------------
f at the beginning of new iteration,        20.1267924183
Predicted improvement:        0.033038060
lambda =          1; f =           20.0860366
Norm of dx   0.017257
----
Improvement on iteration 8 =        0.040755787
-----------------
f at the beginning of new iteration,        20.0860366310
Predicted improvement:        0.002847112
lambda =          1; f =           20.0829777
Norm of dx  0.0065841
----
Improvement on iteration 9 =        0.003058914
-----------------
f at the beginning of new iteration,        20.0829777167
Correct for low angle: 0.00492379
Predicted improvement:        0.000063572
lambda =          1; f =           20.0828715
lambda =     1.9332; f =           20.0828101
Norm of dx 0.00063724
----
Improvement on iteration 10 =        0.000167567
-----------------
f at the beginning of new iteration,        20.0828101497
Predicted improvement:        0.000146558
lambda =          1; f =           20.0825432
lambda =     1.9332; f =           20.0823412
lambda =     3.7372; f =           20.0820797
Norm of dx 0.00071428
----
Improvement on iteration 11 =        0.000730449
-----------------
f at the beginning of new iteration,        20.0820797007
Predicted improvement:        0.000762060
lambda =          1; f =           20.0806480
lambda =     1.9332; f =           20.0794784
lambda =     3.7372; f =           20.0776728
lambda =     7.2247; f =           20.0758836
Norm of dx   0.001488
----
Improvement on iteration 12 =        0.006196116
-----------------
f at the beginning of new iteration,        20.0758835846
Predicted improvement:        0.005621531
lambda =          1; f =           20.0647394
lambda =     1.9332; f =           20.0545177
lambda =     3.7372; f =           20.0352429
lambda =     7.2247; f =           19.9997899
lambda =     13.967; f =           19.9379721
lambda =         27; f =           19.8433630
```

```
lambda =     52.196; f =           19.7528385
Norm of dx  0.0046825
----
Improvement on iteration 13 =        0.323045047
-----------------
f at the beginning of new iteration,        19.7528385381
Predicted improvement:        0.333655599
lambda =          1; f =           19.0853262
lambda =     1.9332; f =           18.4611016
lambda =     3.7372; f =           17.2464071
lambda =     7.2247; f =           14.8471010
lambda =     13.967; f =            9.9055522
lambda =         27; f =           -1.3423529
lambda =     52.196; f =          -29.0329968
lambda =      100.9; f =           20.0956964
lambda =     67.943; f =          638.4403328
lambda =      53.59; f =          -29.8296623
lambda =     61.791; f =           -4.5527269
Norm of dx    0.25108
----
Improvement on iteration 14 =       49.582500870
warning: possible inaccuracy in H matrix
-----------------
f at the beginning of new iteration,       -29.8296623314
Predicted improvement:       28.062951749
lambda =          1; f =          -26.5473936
lambda =    0.33333; f =          -29.6837883
lambda =    0.11111; f =          145.0655790
lambda =   0.037037; f =          -28.3223761
lambda =   0.012346; f =          -30.2371475
Norm of dx     28.071
----
Improvement on iteration 15 =        0.407485194
-----------------
f at the beginning of new iteration,       -30.2371475254
Predicted improvement:        3.080550848
lambda =          1; f =          -35.1852375
lambda =     1.9332; f =          -37.8721471
Norm of dx    0.44125
----
Improvement on iteration 16 =        7.634999607
-----------------
f at the beginning of new iteration,       -37.8721471325
Predicted improvement:        1.595117355
lambda =          1; f =          -39.6787555
Norm of dx     0.2361
----
Improvement on iteration 17 =        1.806608402
-----------------
f at the beginning of new iteration,       -39.6787555343
Predicted improvement:        0.089257596
lambda =          1; f =          -39.7496576
Norm of dx    0.16135
----
Improvement on iteration 18 =        0.070902065
-----------------
f at the beginning of new iteration,       -39.7496575996
Predicted improvement:        0.010710837
```

```
lambda =          1; f =          -39.7615181
Norm of dx   0.039983
----
Improvement on iteration 19 =        0.011860515
-----------------
f at the beginning of new iteration,       -39.7615181148
Predicted improvement:        0.000579287
lambda =          1; f =          -39.7622011
Norm of dx  0.0026865
----
Improvement on iteration 20 =        0.000682964
-----------------
f at the beginning of new iteration,       -39.7622010790
Predicted improvement:        0.000066952
lambda =          1; f =          -39.7623105
lambda =     1.9332; f =          -39.7623676
Norm of dx  0.0013498
----
Improvement on iteration 21 =        0.000166489
-----------------
f at the beginning of new iteration,       -39.7623675679
Predicted improvement:        0.000124273
lambda =          1; f =          -39.7626023
lambda =     1.9332; f =          -39.7627959
lambda =     3.7372; f =          -39.7631006
lambda =     7.2247; f =          -39.7634301
Norm of dx 0.00094306
----
Improvement on iteration 22 =        0.001062503
-----------------
f at the beginning of new iteration,       -39.7634300713
Predicted improvement:        0.001002408
lambda =          1; f =          -39.7653990
lambda =     1.9332; f =          -39.7671710
lambda =     3.7372; f =          -39.7704177
lambda =     7.2247; f =          -39.7760282
lambda =     13.967; f =          -39.7844018
lambda =         27; f =          -39.7914635
Norm of dx  0.0013029
----
Improvement on iteration 23 =        0.028033398
-----------------
f at the beginning of new iteration,       -39.7914634689
Predicted improvement:        0.026431201
lambda =          1; f =          -39.8441674
lambda =     1.9332; f =          -39.8930619
lambda =     3.7372; f =          -39.9867949
lambda =     7.2247; f =          -40.1650430
lambda =     13.967; f =          -40.4985298
lambda =         27; f =          -41.1012367
lambda =     52.196; f =          -42.1041375
lambda =      100.9; f =          -43.3622885
Norm of dx   0.058566
----
Improvement on iteration 24 =        3.570825069
-----------------
f at the beginning of new iteration,       -43.3622885378
Correct for low angle: 0.0038482
```

```
Predicted improvement:        2.599357892
lambda =          1; f =          -46.3818772
Norm of dx     5.8177
----
Improvement on iteration 25 =        3.019588684
-----------------
f at the beginning of new iteration,       -46.3818772214
Correct for low angle: 0.00334657
Predicted improvement:        0.610783035
lambda =          1; f =          -46.2327848
lambda =    0.33333; f =          -46.6191635
Norm of dx     2.6016
----
Improvement on iteration 26 =        0.237286303
-----------------
f at the beginning of new iteration,       -46.6191635240
Predicted improvement:        0.122222278
lambda =          1; f =          -46.7128938
Norm of dx     1.9826
----
Improvement on iteration 27 =        0.093730313
-----------------
f at the beginning of new iteration,       -46.7128938369
Predicted improvement:        0.061499559
lambda =          1; f =          -46.5905380
lambda =    0.33333; f =          -46.7268808
Norm of dx    0.19755
----
Improvement on iteration 28 =        0.013986983
-----------------
f at the beginning of new iteration,       -46.7268808202
Correct for low angle: 0.0048893
Predicted improvement:        0.005121271
lambda =          1; f =          -46.7311180
Norm of dx    0.31569
----
Improvement on iteration 29 =        0.004237165
-----------------
f at the beginning of new iteration,       -46.7311179851
Correct for low angle: 0.00335719
Predicted improvement:        0.000238199
lambda =          1; f =          -46.7311932
lambda =    0.33333; f =          -46.7312314
lambda =    0.64439; f =          -46.7312576
Norm of dx   0.039317
----
Improvement on iteration 30 =        0.000139644
-----------------
f at the beginning of new iteration,       -46.7312576296
Correct for low angle: 0.00333571
Predicted improvement:        0.000025387
lambda =          1; f =          -46.7312673
lambda =    0.33333; f =          -46.7312702
lambda =    0.64439; f =          -46.7312735
Norm of dx   0.012351
----
Improvement on iteration 31 =        0.000015832
-----------------
```

```
f at the beginning of new iteration,       -46.7312734615
Correct for low angle: 0.00311475
Predicted improvement:        0.000003463
lambda =          1; f =          -46.7312733
lambda =    0.33333; f =          -46.7312749
Norm of dx  0.0053369
----
Improvement on iteration 32 =        0.000001444
-----------------
f at the beginning of new iteration,       -46.7312749057
Predicted improvement:        0.000000509
lambda =          1; f =          -46.7312755
Norm of dx  0.0030026
----
Improvement on iteration 33 =        0.000000623
-----------------
f at the beginning of new iteration,       -46.7312755291
Correct for low angle: 0.0036548
Predicted improvement:        0.000000009
lambda =          1; f =          -46.7312755
lambda =    0.33333; f =          -46.7312755
Norm of dx 0.00039814
----
Improvement on iteration 34 =        0.000000002
improvement < crit termination

Final value of minus the log posterior (or likelihood):-46.731276

MODE CHECK

Fval obtained by the optimization routine: -46.731276

RESULTS FROM POSTERIOR ESTIMATION
parameters
        prior mean     mode    s.d.  prior pstdev

gamma_q     0.0000   0.5522  3.7996   norm 5.0000
gamma_N     0.0000   0.6190  0.0776   norm 5.0000

standard deviation of shocks
        prior mean     mode    s.d.  prior pstdev

ep_N        0.0100   0.0520  0.0049   invg 0.0050
ep_Y        0.0100   0.2693  0.0253   invg 0.0050


Log data density [Laplace approximation] is 40.122337.

Estimation::mcmc: Multiple chains mode.
Estimation::mcmc: Old metropolis.log file successfully erased!
Estimation::mcmc: Creation of a new metropolis.log file.
Estimation::mcmc: Searching for initial values...
Estimation::mcmc: Initial values found!

Estimation::mcmc: Write details about the MCMC... Ok!
Estimation::mcmc: Details about the MCMC are available in mcmc/metropolis\mcmc_mh_history_0.mat

,
```

```
Estimation::mcmc: Number of mh files: 1 per block.
Estimation::mcmc: Total number of generated files: 2.
Estimation::mcmc: Total number of iterations: 125000.
Estimation::mcmc: Current acceptance ratio per chain:
                                                       Chain  1: 45.788%
                                                       Chain  2: 45.5536%
Estimation::mcmc: Total number of MH draws per chain: 125000.
Estimation::mcmc: Total number of generated MH files: 1.
Estimation::mcmc: I'll use mh-files 1 to 1.
Estimation::mcmc: In MH-file number 1 I'll start at line 25001.
Estimation::mcmc: Finally I keep 100000 draws per chain.



MCMC Inefficiency factors per block
Parameter       Block 1     Block 2
SE_ep_N          22.573      19.588
SE_ep_Y          23.315      26.275
gamma_q          20.381      16.391
gamma_N          26.534      28.596



Convergence diagnostics results for chain 1.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 1.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_ep_N                0.053            0.005            0.000            0.460
0.437            0.425
SE_ep_Y                0.277            0.027            0.000            0.400
0.408            0.384
gamma_q                2.134            3.406            0.968            0.993
0.992            0.991
gamma_N                0.646            0.093            0.000            0.226
0.251            0.272




Convergence diagnostics results for chain 2.

Geweke (1992) Convergence Tests, based on means of draws 25000 to 45000 vs 75000 to 125000 for
chain 2.
p-values are for Chi2-test for equality of means.
Parameter         Post. Mean        Post. Std   p-val No Taper   p-val 4% Taper   p-val 8% Taper
p-val 15% Taper
SE_ep_N                0.053            0.005            0.025            0.591
0.542            0.505
SE_ep_Y                0.277            0.026            0.395            0.878
0.875            0.854
gamma_q                2.166            3.429            0.000            0.229
0.220            0.176
gamma_N                0.646            0.094            0.000            0.268
0.231            0.190
```

```
Univariate convergence diagnostic, Brooks and Gelman (1998):
    Parameter 1...  Done!
    Parameter 2...  Done!
    Parameter 3...  Done!
    Parameter 4...  Done!
marginal density: I'm computing the posterior mean and covariance...  Done!
marginal density: I'm computing the posterior log marginal density (modified harmonic mean)...
Done!


ESTIMATION RESULTS

Log data density (Modified Harmonic Mean) is 40.139983.
parameters
          prior mean   post. mean        90% HPD interval    prior       pstdev
gamma_q        0.000       2.1298     -3.8957      7.3179     norm       5.0000
gamma_N        0.000       0.6461      0.5006      0.7979     norm       5.0000

standard deviation of shocks
          prior mean   post. mean        90% HPD interval    prior       pstdev

ep_N           0.010       0.0534      0.0452      0.0618     invg       0.0050
ep_Y           0.010       0.2772      0.2333      0.3191     invg       0.0050
--- Parameters Post. Std ---
gamma_q: 3.413351
gamma_N: 0.092732
--- Shocks Post. Std ---
ep_N: 0.005180
ep_Y: 0.026622
Total computing time : 0h22m52s
>>
```